

ROS Installation

There are several options when installing ROS on your PC. These options range from installations designed for embedded PCs to full-fledged development machines. Let's go through each of them.

ROS Base

It installs ROS package, build and communication libraries.

It is recommended for embedded computers, where you don't have much hardware for graphical tools or real robots (in the production stage) to run their specific tasks, without the need of debugging.

- `sudo apt install ros-melodic-ros-base`

ROS Desktop

It installs ROS Base, rqt, rviz and robot-generic libraries.

These are the basic packages for ROS development. It's recommended for beginners to follow tutorials about ROS without having to install too much. RQT and RVIZ provides graphical interfaces for visualizing what is happening behind the scenes. With this option, you don't have to work only using shell.

- `sudo apt install ros-melodic-desktop`

ROS Desktop Full

It installs everything of ROS Desktop option plus 2D/3D simulators and 2D/3D perception.

If you want to simulate using ROS default Gazebo version, that's your option. The Gazebo versions that comes with ROS Melodic is 9.0.

- `sudo apt install ros-melodic-desktop-full`

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For convenience, it's best that you be installing the ROS Base option, because it is the quickest process. After running the command, you must have confirmation at the end of the output in your terminal:

- 0 upgraded, 485 newly installed, 0 to remove and 3 not upgraded.
- Need to get 234 MB of archives.